

ARYAN SACHDEVA

Roorkee, Uttarakhand

+91-8267829489

aryan01001s@gmail.com

[Portfolio](#)

[LinkedIn](#)

[GitHub](#)

[LeetCode](#)

EDUCATION

Indian Institute of Technology, Jodhpur

Aug 2024 – Apr 2028

Bachelor's Degree - Chemical Engineering - CGPA - 8.03

Jodhpur, Rajasthan

TECHNICAL SKILLS

Languages: Python, C, C++, SQL, JavaScript

Backend: Django, Django REST Framework, FastAPI, REST APIs, Celery, Async Task Processing

Databases & Search: PostgreSQL, Redis, FAISS, Vector Search, BM25

AI/ML & Libraries: PyTorch, TensorFlow, Scikit-learn, LangChain, RAG, LLMs, Sentence Transformers

DevOps & Tools: Docker, Git, GitHub, GitHub Actions, CI/CD, Nginx, AWS, Render, Vercel

Core Concepts: Data Structures, Algorithms, OOP, Computer Networks, System Design

EXPERIENCE

PHC Booking System | [GitHub](#) | *React, Django, PostgreSQL, JWT*

Jan 2026 – Apr 2026

Design Credit Project under Prof. Romi Banerjee

Indian Institute of Technology, Jodhpur

- Architected a React and Django REST Framework healthcare booking platform supporting 3 user roles and slot-management workflows.
- Prevented double-booking under concurrent requests with atomic transactions and row-level locking using `select_for_update()`.
- Optimized PostgreSQL query performance 50%+ with `select_related()` and `prefetch_related()`, eliminating N+1 queries.
- Secured 10+ API endpoints with JWT authentication, RBAC permission classes, and Axios-based silent token refresh.

PROJECTS

TaskForge | [GitHub](#) | *FastAPI, Celery, Redis, PostgreSQL, React, Docker*

May 2026 – Jun 2026

- Developed an async image-processing platform with FastAPI, Celery, Redis, and PostgreSQL.
- Containerized 5 services behind Nginx with Docker Compose and healthcheck-gated startup, with API responses under 150ms.
- Engineered 2 isolated Celery priority queues with dedicated worker pools, completing high-priority jobs in under 1s despite a 10-job backlog.
- Implemented exponential-backoff retries (1s/2s/4s), dead-letter recovery, and idempotency guards against duplicate task delivery.

Knowledge Base | [Live](#) | [GitHub](#) | *FastAPI, React, FAISS, BM25, Groq*

Oct 2025 – Nov 2025

- Constructed hybrid RAG with FAISS, BM25, and Reciprocal Rank Fusion (k=60) to retrieve top 5 chunks from multi-document corpora.
- Exposed 8 FastAPI REST/SSE endpoints for PDF ingestion and Groq token streaming at sub-2s latency.
- Persisted FAISS indexes with 384-dim ONNX embeddings and deployed the React frontend on Vercel with FastAPI backend on Hugging Face.

COURSEWORK

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|--------------------|------------------------------|-----------------------|
| • Data Structures | • Intro to Computer Science | • Linear Algebra |
| • Machine Learning | • Probability and Statistics | • Signals and Systems |

SCHOLASTIC ACHIEVEMENTS

- Solved **500+** algorithmic problems across LeetCode, Codeforces, and GeeksforGeeks.
- Earned **A** grades in **Data Structures** and **Machine Learning** coursework.

LEADERSHIP

- Led marketing outreach for **Prometeo** using sponsor coordination and volunteer-driven promotions, handling **3 sponsors** with a team of **10+ volunteers**.
- Coordinated informal events for **Ignus** using logistics planning and on-ground volunteer execution, managing participant flow across **5 events**.
- Managed event operations for **SandStone** using scheduling, speaker coordination, and team execution, supporting **3 speakers**.